T5

$\xi \sim R[\theta, 2\theta] \Rightarrow \rho(x) = \dfrac{1}{\theta}, \quad \theta > 0$

a) $\vec{X}_n$ - выборка

1) $\alpha_1 = \int\limits_{-\infty}^{+\infty} x \rho(x) dx = \int\limits_{\theta}^{2\theta} \dfrac{x}{\theta} dx = \dfrac{x^2}{2\theta}\Big|_{\theta}^{2\theta} = \dfrac{4\theta^2}{2\theta} - \dfrac{\theta^2}{2\theta} = \dfrac{3\theta}{2} = \alpha_1(\theta)$

$\alpha_1(\tilde{\theta}) = \tilde{\alpha}_1 \rightarrow \frac{3}{2}\tilde{\theta} = \bar{X} \Rightarrow \boxed{\tilde{\theta}_1 = \frac{2}{3}\bar{X}}$ ОММ

2) $L(\theta) = \prod\limits_{i=1}^{n} \rho(x_i; \theta) = \dfrac{1}{\theta^n}$ убывает по θ

$\theta \leq x_{min} \qquad x_{max} \leq 2\theta \Rightarrow \dfrac{x_{max}}{2} \leq \theta \leq x_{min} \rightarrow \sup L = \theta_{min} = \dfrac{x_{max}}{2} \rightarrow \boxed{\tilde{\theta}_2 = \dfrac{x_{max}}{2}}$ ОМП

b) 1) $M\tilde{\theta} = \theta$ - несмещённость

$\frac{2}{3} M \frac{1}{n}\sum\limits_{i=1}^{n} x_i = \frac{2}{3} M\xi = \frac{2}{3} \cdot \frac{3}{2}\theta = \theta \rightarrow$ оценка ОММ несмещённая

т.к. оценка $\tilde{\theta}_1$ несмещённая проверим достаточное условие

$M\xi^2 = \int\limits_{\theta}^{2\theta} x^2 \frac{1}{\theta} dx = \frac{x^3}{3\theta}\Big|_{\theta}^{2\theta} = \frac{8\theta^3}{3\theta} - \frac{\theta^3}{3\theta} = \frac{7}{3}\theta^2; \quad D\xi = M\xi^2 - (M\xi)^2 = \frac{7}{3}\theta^2 - \frac{9}{4}\theta^2 = \frac{1}{12}\theta^2$

$D\tilde{\theta}_1 = D\left[\frac{2}{3}\frac{1}{n}\sum x_i\right] = \frac{4}{9}\frac{1}{n^2} n D\xi = \frac{4}{9n} \cdot \frac{\theta^2}{12} \xrightarrow[n \to \infty]{} 0 \Rightarrow \tilde{\theta}_1$ – состоятельная

2) $M\tilde{\theta}_2 = \frac{1}{2} M x_{max} = \frac{1}{2} M\xi_{(n)}$

$\xi_{(n)} \sim (F(x))^n$

$\rho_{(n)}(x) = \left((F(x))^n\right)' = n F(x)^{n-1} \cdot \rho(x) = \dfrac{n}{\theta} \cdot F(x)^{n-1},$

$F^{n-1}(x) = \begin{cases} (x/\theta - 1)^{n-1}, & x \in [\theta, 2\theta] \\ 1, & x \geq 2\theta \\ 0, & x \leq \theta \end{cases}$

$M\xi_{(n)} = \int\limits_{\theta}^{2\theta} x \frac{n}{\theta}\left(\frac{x}{\theta} - 1\right)^{n-1} dx = \left\{ \begin{array}{l} \frac{x}{\theta} - 1 = t \\ x = \theta(t+1) \\ dx = \theta dt \end{array} \right\} = \frac{n}{\theta}\int\limits_{0}^{1} \theta^2 (t+1) t^{n-1} dt =$

$= n\theta \int\limits_{0}^{1} t^n dt + n\theta \int\limits_{0}^{1} t^{n-1} dt = \dfrac{n\theta}{n+1} + \theta = \dfrac{n\theta + n\theta + \theta}{n+1} = \dfrac{\theta(2n+1)}{n+1}$

$\Rightarrow M\tilde{\theta}_2 = \dfrac{\theta(2n+1)}{2(n+1)} \neq \theta \Rightarrow \tilde{\theta}_2$ смещённая оценка

$\boxed{\tilde{\theta}_2' = \dfrac{2(n+1)\tilde{\theta}_2}{(2n+1)} \text{ – исправленная оценка}}$

$$M\xi_{(n)}^2 = \int_\theta^{2\theta} x^2 \frac{n}{\theta}\left(\frac{x}{\theta}-1\right)^{n-1} dx = \left\{\begin{matrix} \frac{x}{\theta}-1=t \\ x=\theta(t+1) \\ dx=\theta dt \end{matrix}\right\} = \int_0^1 \theta^3 (t+1)^2 \frac{n}{\theta} t^{n-1} dt =$$

$$= n\theta^2 \int_0^1 \left[t^2+2t+1\right] t^{n-1} dt = n\theta^2 \int_0^1 t^{n+1} dt + 2n\theta^2 \int_0^1 t^n dt + n\theta^2 \int_0^1 t^{n-1} dt =$$

$$= \frac{n\theta^2}{n+2} + \frac{2n\theta^2}{n+1} + \frac{\cancel{n}\theta^2}{\cancel{n}}$$

$$D\xi_{(n)} = M\xi_{(n)}^2 - \left(M\xi_{(n)}\right)^2 = \frac{n\theta^2}{n+2} + \frac{2n\theta^2}{n+1} + \theta^2 - \frac{\theta^2(2n+1)^2}{(n+1)^2} =$$

$$= \theta^2 \frac{(n+1)^2 n + 2n(n+1)(n+2) + (n+1)^2(n+2) - (2n+1)^2(n+2)}{(n+2)(n+1)^2} =$$

$$= \theta^2 \frac{\cancel{n^3} + \cancel{2n^2} + n + \cancel{2n^3} + \cancel{6n^2} + \cancel{4n} + \cancel{n^3} + \cancel{4n^2} + \cancel{5n} + \cancel{2} - \cancel{4n^3} - \cancel{8n^2} - \cancel{4n^2} - \cancel{9n} - \cancel{2}}{(n+2)(n+1)^2}$$

$$= \theta^2 \frac{n}{(n+2)(n+1)^2}$$

проверим несмещённую оценку по достаточному условию

$$D\tilde{\theta}_2' = \frac{(n+1)^2}{(2n+1)^2} D\xi_{(n)} = \frac{\cancel{(n+1)^2}}{(2n+1)^2} \theta^2 \frac{n}{\cancel{(n+1)^2}(n+2)} \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_2' \text{ — состоят.}$$

c) неравенством Крамера-Рао нельзя пользоваться, т.к. модель нерегулярная

$$D\tilde{\theta}_1 = \frac{4}{9n} \cdot \frac{\theta^2}{12} \quad > \quad D\tilde{\theta}_2' = \frac{\theta^2 n}{(2n+1)^2(n+2)}$$

$\Rightarrow \tilde{\theta}_2'$ эффективнее, чем $\tilde{\theta}_1$

d)

$$\eta = \frac{\xi-\theta}{\theta} \sim R(0,1)$$

$$\eta_{max} = \frac{X_{max}-\theta}{\theta} \sim \begin{cases} 1, & x \geq 1 \\ x^n, & x \in (0,1) \\ 0, & x \leq 0 \end{cases}$$

$$t_1 = q_{\frac{1-\beta}{2}} \Rightarrow t_1^n = \frac{1-\beta}{2}$$

$$t_1 = \left(\frac{1-\beta}{2}\right)^{1/n}$$

$$P\left(t_1 < \frac{X_{max}}{\theta} - 1 < t_2\right) = \beta$$ аналогично $t_2 = \left(\frac{1+\beta}{2}\right)^{1/n}$

$$t_1 + 1 < \frac{X_{max}}{\theta} < t_2 + 1$$

$$\boxed{\frac{X_{max}}{t_2+1} < \theta < \frac{X_{max}}{t_1+1}}$$ точный доверительный интервал

е) 1) $\tilde{\theta}_1 = \frac{2}{3}\tilde{\alpha}_1$; $\tilde{\theta}_1 = g(\tilde{\alpha})$; $g(\alpha) = \theta$

$g(\alpha) \in C(\alpha)$, $\nabla g(\alpha) = 3/2 \neq \vec{0}$ $\Rightarrow$ можем применить Ⓣ

$$\sigma^2(\alpha) = \nabla^T g(\alpha)\, K\, \nabla g(\alpha) \qquad K = \alpha_2 - \alpha_1^2$$

$$\sigma^2(\alpha) = \left(3/2\right)^2\left(\alpha_2 - \alpha_1^2\right) = \frac{9}{4}\left(\alpha_2 - \alpha_1^2\right)$$

$$\sqrt{n}\,\frac{\tilde{\theta} - \theta}{3/2\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} \rightsquigarrow N(0,1)$$

$$P\left(t_1 < \sqrt{n}\,\frac{\tilde{\theta} - \theta}{3/2\sqrt{\tilde{\alpha}_2 - \tilde{\alpha}_1^2}} < t_2\right) = \beta, \quad t_1 = N_{\frac{1-\beta}{2}}(0,1)\ ; \quad t_2 = N_{\frac{1+\beta}{2}}(0,1)$$

$$\underbrace{\tilde{\theta} - \frac{3/2\sqrt{\dots}}{\sqrt{n}}t_2 < \theta < \tilde{\theta} - \frac{3/2\sqrt{\dots}}{\sqrt{n}}t_1}$$ асимптотический доверительный интервал по ОММ

2) асимптотический доверительный интервал по ОМП построить не можем, т.к. для выполнения Ⓣ нужна сильно регулярная модель, но наша даже не регулярна

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Т6

$$\xi \sim p(x) = \begin{cases} \frac{\theta - 1}{x^\theta}, & x \geq 1 \\ 0, & x < 1 \end{cases} \qquad \theta > 1 \qquad \vec{X}_n \text{ — выборка}$$

a) $L(\theta) = \prod_{i=1}^{n} p(x_i, \theta) = \dfrac{(\theta-1)^n}{\left(\prod_{i=1}^{n} x_i\right)^{\theta}}$

$\ln L(\theta) = n \ln(\theta-1) - \theta \sum \ln x_i$

$\ln L(\theta)' = \dfrac{n}{\theta-1} - \sum \ln x_i = 0 \Rightarrow \dfrac{n - \sum \ln x_i (\theta-1)}{\theta-1} = 0 \Rightarrow \boxed{\tilde{\theta} = \dfrac{n}{\sum \ln x_i} + 1}$ ОМП

$\ln L(\theta)'' = -\dfrac{n}{(\theta-1)^2}\Big|_{\tilde{\theta}} < 0 \Rightarrow$ т. max

c) для начала проверим является ли распределение Парето сильно регулярным.

1) $F(x) = \int_{-\infty}^{+\infty} p(x) dx \in C''_{\theta}(\Theta)$

2) $I(\theta) = \int_{A(\theta)} \left(\dfrac{\partial \ln p(x,\theta)}{\partial \theta}\right)^2 p(x,\theta) dx = \int_1^{+\infty} \left[\dfrac{\partial(\ln(\theta-1) - \theta \ln x)}{\partial \theta}\right]^2 \dfrac{\theta-1}{x^{\theta}} dx =$

$= \int_1^{+\infty} \left[\dfrac{1}{(\theta-1)^2} + \ln^2 x - \dfrac{2\ln x}{\theta-1}\right] \dfrac{\theta-1}{x^{\theta}} dx = \int_1^{+\infty} \left(\dfrac{1}{(\theta-1)x^{\theta}} + \dfrac{\ln^2 x(\theta-1)}{x^{\theta}} - \dfrac{2\ln x}{x^{\theta}}\right) dx =$

$= \left\{\begin{aligned} &\frac{1}{x^{\theta}} = t \\ &x = \sqrt[\theta]{\frac{1}{t}} = t^{-1/\theta} \\ &dx = -\frac{1}{\theta} t^{-\frac{\theta+1}{\theta}} dt \\ &\ln x = -1/\theta \ln t \end{aligned}\right\} = \int_0^1 \dfrac{t^{-\frac{\theta+1}{\theta}}}{\theta t(\theta-1)} dt + \int_0^1 \dfrac{t \ln^2 t(\theta-1) \cdot t^{-\frac{\theta+1}{\theta}}}{\theta^3} dt + \int_0^1 \dfrac{2\ln t \cdot t \cdot t^{-\frac{\theta+1}{\theta}}}{\theta^2} dt$

$= \int_0^1 \dfrac{t^{\frac{-2\theta-1}{\theta}}}{\theta(\theta-1)} dt + \int_0^1 \dfrac{(\theta-1)\, t^{-1/\theta} \ln^2(t)}{\theta^3} dt + \int_0^1 \dfrac{2\, t^{-1/\theta} \ln t}{\theta^2} dt = \dfrac{1}{(1-\theta)^2}$

$\dfrac{-2}{(\theta-1)^2}$

$$\frac{1}{(1-\theta)^2} \qquad \frac{2}{(\theta-1)^2}\int_0^1 t^{-1/\theta}\ln^2 t\,dt = \frac{\theta}{\theta-1}\int_0^1 \ln^2 t\,d\,t^{\frac{\theta-1}{\theta}} =$$

$$= \frac{\theta}{\theta-1}\ln^2 t\cdot t^{\frac{\theta-1}{\theta}}\Big|_0^1 - \frac{\theta}{\theta-1}\int_0^1 2\cdot t^{\frac{\theta-1}{\theta}-1}\ln t\,dt =$$

$$= -\frac{\theta^2}{(\theta-1)^3}\int_0^1 2\cdot\ln t\,d\,t^{\frac{\theta-1}{\theta}} = -\frac{(\theta)^2}{(\theta-1)^3}2\ln t\cdot t^{\frac{\theta-1}{\theta}}\Big|_0^1 +$$

$$+\frac{\theta^2}{(\theta-1)^3}\int_0^1 t^{\theta-1/\theta-1}\,dt = \frac{2\theta}{(\theta-1)^2}$$

$I(\theta)\in C_\theta(\Theta)$; $I(\theta)>0$

3) $$\int_{-\infty}^{+\infty} d\,\frac{\partial F(x,\theta)}{\partial\theta} = 0$$

$$\int_1^{+\infty}\frac{\partial p(x,\theta)}{\partial\theta}dx = \int_1^{+\infty}\left(\frac{1}{x^\theta} - \frac{(\theta-1)\ln x}{x^\theta}\right)dx = \frac{1}{\theta-1} - (\theta-1)\cdot\frac{1}{(\theta-1)^2} = 0 \quad \checkmark$$

$$\int_1^{+\infty}\frac{\partial^2 p(x,\theta)}{\partial\theta^2}dx = \int_1^{+\infty}\left[-\frac{\ln x}{x^\theta} - \frac{\ln x}{x^\theta} + \frac{(\theta-1)\ln^2 x}{x^\theta}\right]dx = 0 \quad \checkmark$$

$\Rightarrow$ модель сильно регулярная

$\Rightarrow$ можно применять (T) для асимптотического интервала ОМП.

$g(\theta)=\theta$; $\quad g(\tilde\theta)=\tilde\theta$

$$\sigma^2(\theta) = \nabla^T g(\theta)\, I^{-1}(\theta)\,\nabla g(\theta) = (1-\theta)^2$$

$$\Rightarrow \sqrt{n}\,\frac{\tilde{\theta}-\theta}{1-\tilde{\theta}} \sim N(0,1)$$

$t_1 = -1{,}96$

$t_2 = 1{,}96.$

$$P\left(t_1 < \sqrt{n}\,\frac{\frac{n}{\sum \ln x_i} - 1 - \theta}{1-\tilde{\theta}} < t_2\right) = \beta$$

$$\underbrace{\tilde{\theta} - \frac{t_1(1-\tilde{\theta})}{\sqrt{n}} < \theta < \tilde{\theta} - \frac{t_2(1-\tilde{\theta})}{\sqrt{n}}}$$

асимптотичний доверит. интервал

b) q_2 - медиана $\Rightarrow F(x) = 1/2$

$$F(x) = \int_1^x \frac{\theta-1}{t^\theta}\,dt = -\frac{(\theta-1)}{\theta-1}\left(x^{\frac{1}{\theta-1}} - 1\right) = 1 - 1/x^{\theta-1}$$

$1 - x^{1-\theta} = 1/2 \Rightarrow x = 2^{1/\theta-1}$ - медиана

$q_2 \sim Bi(n, 1/2)$

$P\left(X_{(t_1)} < q_2 < X_{(t_2)}\right) = \beta$

$t_1 = Bi_{\frac{1-\beta}{2}}(n, 1/2)$

$t_2 = Bi_{\frac{1+\beta}{2}}(n, 1/2)$

$$\underbrace{X_{(t_1)} < q_2 < X_{(t_2)}}$$

точный доверительный интервал для медианы